DUNMAN SECONDARY SCHOOL

CANDIDATE NAME			
CLASS		**INDEX NUMBER**	

PRELIMINARY EXAMINATION 2023
SECONDARY 4 EXPRESS

ADDITIONAL MATHEMATICS **4049/02**

Paper 2 28 August 2023

2 hours 15 minutes

Marking Scheme

(Updated 1 Dec 2022 by HOD/Math aligned to Marking Scheme for Specimen Papers 4049)

Marking Scheme for 2023 Prelim Add Math Paper 2

Qn	Solution	Marks / Comments
1(a)	$\frac{d}{dx} = 2e^{2x+1}(4\cos 4x) + \sin 4x(2(2)e^{2x+1})$ $= 4e^{2x+1}\left[2\cos 4x + \sin 4x\right]$ *or* $= 8e^{2x+1}\cos 4x + 4e^{2x+1}\sin 4x$	M2 (for differentiation of $2e^{2x+1}$ and $\sin 4x$) A1
1(b)	$y = \frac{x-2}{3x+2}$ $\frac{dy}{dx} = \frac{(3x+2)(1)-(x-2)(3)}{(3x+2)^2}$ $= \frac{8}{(3x+2)^2}$ $For\ stationary\ po\int s, \frac{dy}{dx} = 0$ $\text{Sin}\,ce\ (3x+2) > 0\, and\ hence\, (3x+2)^2 > 0.$ $Therefore\ \frac{8}{(3x+2)^2} > 0.$ Since $\frac{8}{(3x+2)^2} > 0$, there is no turning point.	M1 (quotient rule) M1 A1 (explanation)

2(a)	$x^2+y^2+px+\left(\frac{p}{2}+4\right)y+k=0$ $x^2+y^2+2ax+2by+a^2+b^2-r^2=0$ $2a=p \quad 2b=\frac{p}{2}+4 \quad a^2+b^2-r^2=k$ $a=\frac{p}{2} \quad b=\frac{p}{4}+2$ $Center\ M=(-\frac{p}{2},-(\frac{p}{4}+2))$ $Sub\ M\ to\ eqn,$ $-(\frac{p}{4}+2)=\frac{3}{2}(-\frac{p}{2})-4$ $-\frac{p}{4}-2=-\frac{3p}{4}-4$ $p=-4\ (shown)$ $\therefore M=(-\frac{-4}{2},-(\frac{-4}{4}+2))$ $=(2,-1)$	B1 (recall method to find center) B1 (expression for center) M1 A1 A1(e.c.f)
2(b)	$\text{Since } x=-8 \text{ is tangent, therefore radius}=10$ $a^2+b^2-r^2=k$ $(-2)^2+(1)^2-10^2=k$ $k=-95.$	B1 (radius=10) A1
2(c)	$Dist\ AM=\sqrt{(14-2)^2+(8-(-1))^2}$ $=\sqrt{225}$ $=15$ $\text{Since } AM\ (15) > \text{radius of } 10, \text{therefore point } A \text{ is outside of circle.}$ With Point A outside of circle, then angle XAY must be less than 90° because when it is on the circle, the angle will be 90° (right angle in semi-circle)	M1 B1 (as long as students are able to explain outside of circle and relate to right-angle in semi-circle)
3(a)	$AP=\frac{2}{3}(9)=6cm$ $PQ=2(6\sin\theta)$ $=12\sin\theta$ $PS=3\cos\theta$	M2(each for *PQ* and *PS*)

	$L = 2PQ + 2PS + AC + AB + 2(SB)$ $= 24\sin\theta + 6\cos\theta + 9 + 9 + 2(3\sin\theta)$ $= 18 + 30\sin\theta + 6\cos\theta \ (shown)$	A1
3(b)	$30\sin\theta + 6\cos\theta = R\sin(\theta+\alpha)$ $= R\sin\theta\cos\alpha + R\cos\theta\sin\alpha$ $R\cos\alpha = 30$ $R\sin\alpha = 6$ $\alpha = \tan^{-1}(\frac{1}{5})$ $\quad R^2 = 30^2 + 6^2$ $= 11.31°$ $\quad R = \sqrt{936}$ $= 6\sqrt{26}$ $L = 18 + 6\sqrt{26}\sin(\theta + 11.3°)$	M2 A1 (must simplify to $6\sqrt{26}$)
3(c)	$35 = 18 + 6\sqrt{26}\sin(\theta + 11.31°)$ $\frac{17}{6\sqrt{26}} = \sin(\theta + 11.31°)$ $ref\,(\theta + 11.31°) = 33.756°$ $(\theta + 11.31°) = 33.756°,\ 146.244°$ $\theta = 22.4°, 134.9°(NA)$	M1 M1 A1
4(a)	$y = \ln\left(\frac{\sqrt{x^2+9}}{2x-4}\right)$ $\frac{\sqrt{x^2+9}}{2x-4} > 0\,(to\ be\ defined)$ $\text{Sin}\,ce\ \sqrt{x^2+9} > 0,\ then\ 2x-4 > 0$ $x > 2$ $k = 2$	M1 (state $\frac{\sqrt{x^2+9}}{2x-4} > 0$) A1

4(b)	$y=\ln(\frac{\sqrt{x^2+9}}{2x-4})$ $=\frac{1}{2}\ln(x^2+9)-\ln(2x-4)$ $\frac{dy}{dx}=\frac{1}{2}(\frac{2x}{x^2+9})-\frac{2}{2x-4}$ $=\frac{x}{x^2+9}-\frac{2}{2x-4}$ $=\frac{x(2x-4)-2(x^2+9)}{(x^2+9)(2x-4)}$ $=\frac{-4x-18}{(x^2+9)(2x-4)}$ $\mathrm{Sin}\,ce\ x>2,\ (-4x-18)<0$ $(x^2+9)>0\ and\ (2x-4)>0$ $\therefore \frac{dy}{dx}<0,\ it\,is\ a\ decrea\sin g\ function.$	M1 (simplification) M1, M1 (for each differentiation) M1 (state at least 2 conditions) A1
5(a)	$a=6$ $b=2$ $c=3$	A1 A1 A1
6(b)	$-b=gradient$ $\quad a=2.15$ $=\frac{7-2.8}{3-0.4}$ $b=-1.62$	A1(+/- 0.3) gradient A1 (+/- 0.1) y-intercept
6(c)	$\frac{V}{x}=6,\ from\ graph$ $x=2.35$	M1(using graph for V/x=6) A1 (+/- 0.1)
6(d)	$V=[1.6153(8)+2.15]\times 8$ $=120.57cm^3$ or $from\ graph,$ $\frac{V}{x}=12.5,$ $height=6.1\neq 8(NA)$ Peter is not correct in his estimation.	M1 A1 (statement)
7(a)	Coordinate of Q = (p, $4p^2$) Area of RQS = 0.5 (p) ($4p^2$-2) = $2p^3$- p (shown)	M1 (sub for area of triangle) A1

7(b)	$\frac{dp}{dt}=0.03$ $A=2p^3-p$ $\frac{dA}{dp}=6p^2-1$ $\therefore \frac{dA}{dt}=(6p^2-1)(0.03)$ $=(6(2)^2-1)(0.03)$ $=0.69\ units/s$	M1 M1(connecting rate) A1
8(a)	Perpendicular length of triangle $=\frac{8-2x}{2}=4-x$ Let height of pyramid be *h*. $(4-x)^2=x^2+h^2$ $h^2=16-8x+x^2-x^2$ $h=\sqrt{16-8x}$ $=2\sqrt{4-2x}\ (shown)$	M1 M1(Pythagoras) A1
8(b)	$V=\frac{1}{3}(2x)^2(2\sqrt{4-2x})$ $=\frac{8}{3}x^2\sqrt{4-2x}$ $\frac{dV}{dx}=\frac{8}{3}x^2(\frac{1}{2})(4-2x)^{-\frac{1}{2}}(-2)+(4-2x)^{\frac{1}{2}}(\frac{16}{3}x)$ $=\frac{8}{3}(4-2x)^{-\frac{1}{2}}\left[x^2(-1)+(4-2x)(2x)\right]$ $=\frac{8[-x^2+8x-4x^2]}{3\sqrt{4-2x}}$ *or* $\frac{-8}{3}x(4-2x)^{-\frac{1}{2}}+\frac{16}{3}x(4-2x)^{\frac{1}{2}}$ $=\frac{8(-5x^2+8x)}{3\sqrt{4-2x}}$	M1 M1 (chain rule for $\sqrt{4-2x}$) M1(use of product rule) M1 (equate $\frac{dV}{dx}=0$) M1

	$\frac{dV}{dx}=\frac{8(-5x^2+8x)}{3\sqrt{4-2x}}$ $\frac{8(-5x^2+8x)}{3\sqrt{4-2x}}=0$ $8(-5x^2+8x)=0$ $x=0\,(NA) \quad or \quad x=1.6$ $Considering\ \frac{\mathrm{d}y}{\mathrm{d}x},$ $\begin{array}{c\|c\|c\|c} x & 1 & 1.6 & 1.8 \\ \hline dV/dx & +ve & 0 & -ve \end{array}$ It is a maximum volume, *V*.	A1 (show working to determine nature)

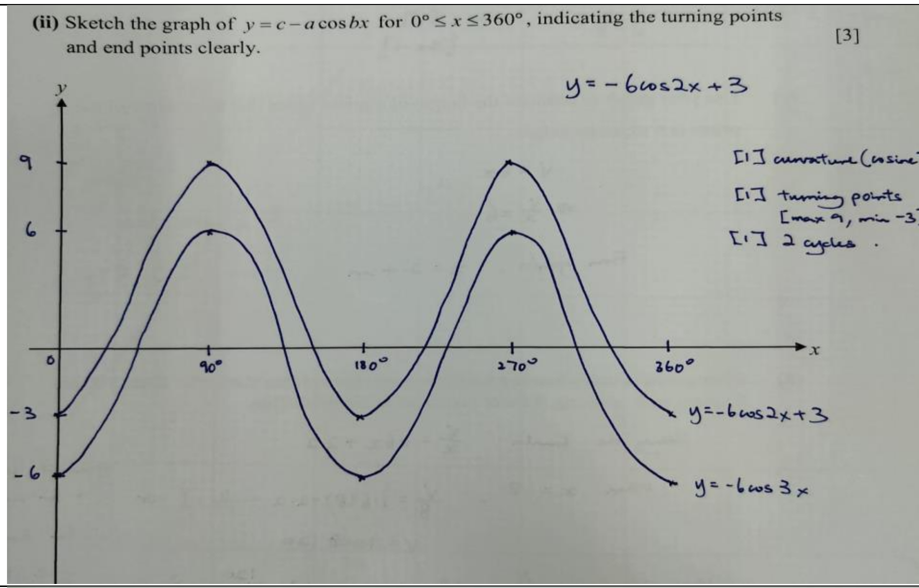
9(a)	$y = 4x + 8 ----(1)$ $y = 2(x-3)^2 + 4 \ ----(2)$ $(1) = (2)$ $4x + 8 = 2(x^2 - 6x + 9) + 4$ $2x^2 - 16x + 14 = 0$ $x = 1,\ 7.$ $Sub\ \ x = 7,$ $y = 36.$ $P(7,\ 36)$	M1(combine equations) M1(quadratic formula to solve) A1
9(b)	$A_1 = \int_0^1 2x^2 - 16x + 14\, dx$ $= \left[\frac{2}{3}x^3 - 8x^2 + 14x \right]_0^1$ $= 6\frac{2}{3}\ units^2$ $A_2 = \int_1^7 [-2x^2 + 16x - 14]\, dx$ $= \left[-\frac{2}{3}x^3 + 8x^2 - 14x \right]_1^7$ $= 72$ $Total\ \ area = 72 + 6\frac{2}{3}$ $= 78\frac{2}{3} units^2$	M1 (or equivalent) A1 M1 (or equivalent) A1 A1
10(a)	$a = \frac{d}{dx}(2t^2 - 9t - 5)$ $= 4t - 9$ $Sub\, t = 1,$ $a = -5\, m/s^2$	M1 A1

10(b)	$s=\int 2t^2-9t-5\,dt$ $=\frac{2}{3}t^3-\frac{9}{2}t^2-5t+c$ $Sub\ t=0,\ s=8$ $8=c.$ $s=\frac{2}{3}t^3-\frac{9}{2}t^2-5t+8$	M1 M1 (sub s=8) A1
10(c)	Instantaneous rest, $v=0$, $2t^2-9t-5=0$ $t=5,-0.5\,(NA)$ Displacement from t = *0* to *t=5*, $Displacement=\left[\frac{2}{3}t^3-\frac{9}{2}t^2-5t\right]_0^5$ $=-\frac{325}{6}$ Displacement from *t=5* to *t=10,* $Displacement=\left[\frac{2}{3}t^3-\frac{9}{2}t^2-5t\right]_5^{10}$ $=\frac{1325}{6}$ Total distance travelled = 275 m	M1 M1 (or equivalent) M1 (or equivalent) A1`

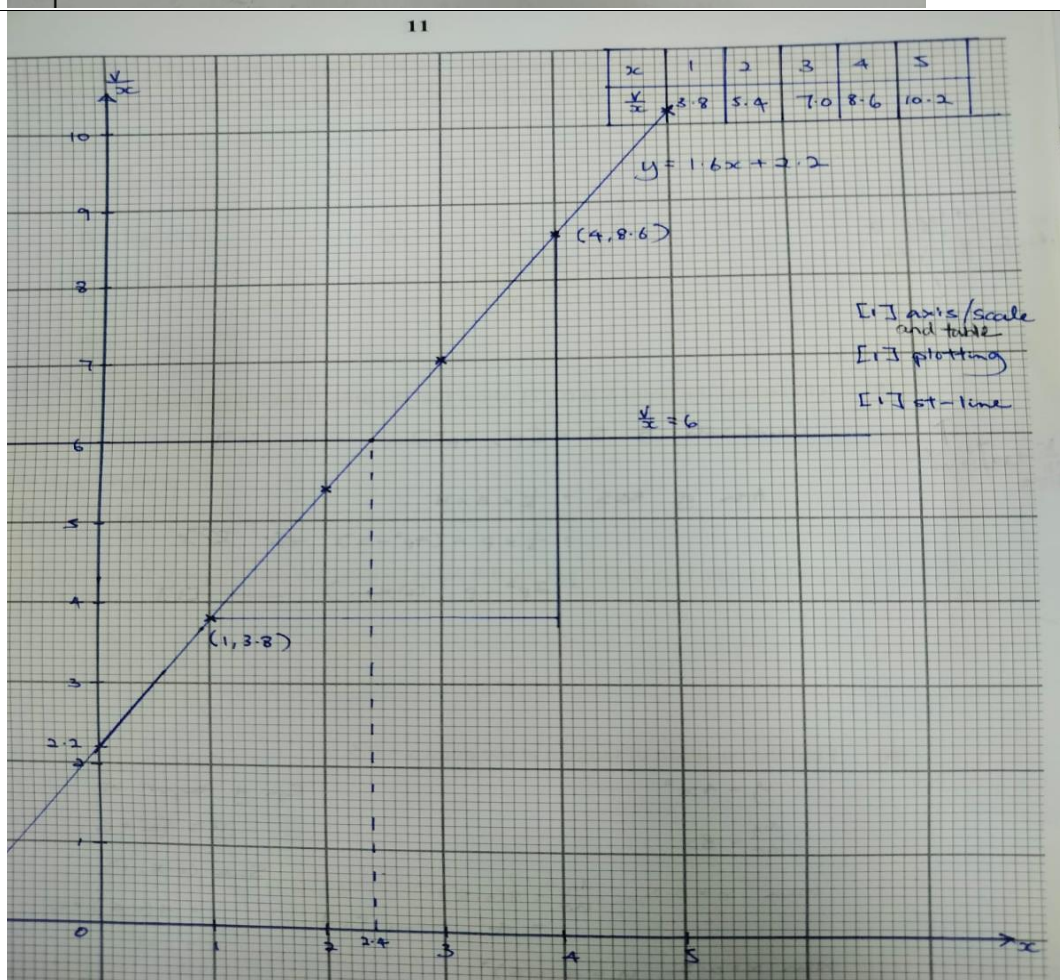
11(a)	$AC = AD\cos 30°$ $= \frac{\sqrt{3}}{2}AD$ $DC = AD\sin 30°$ $= \frac{1}{2}AD$ $\therefore BC = 2DC = AD$ $AB^2 = AC^2 + BC^2$ $= (\frac{\sqrt{3}}{2}AD)^2 + (AD)^2$ $= \frac{7}{4}AD^2$ $AB = \frac{\sqrt{7}}{2}AD\ (shown)$	M1 (either AC or DC shown) M1(Pythagoras) A1 [if students use numbers to represent sides, only give 1 mark unless the explain the representation, let AD be 1u]
11(b)	$\tan(x+30) = \frac{BC}{AC}$ $= \frac{AD}{\left(\frac{\sqrt{3}}{2}AD\right)}$ $= \frac{2}{\sqrt{3}}$ $\frac{\tan x + \tan 30}{1 - \tan x \tan 30} = \frac{2}{\sqrt{3}}$ $\frac{\tan x + \frac{1}{\sqrt{3}}}{1-(\tan x)(\frac{1}{\sqrt{3}})} = \frac{2}{\sqrt{3}}$ $\sqrt{3}\tan x + 1 = 2 - \frac{2}{\sqrt{3}}\tan x$ $3\tan x + \sqrt{3} = 2\sqrt{3} - 2\tan x$ $5\tan x = \sqrt{3}$ $\tan x = \frac{\sqrt{3}}{5}$ $x = \tan^{-1}(\frac{\sqrt{3}}{5})\ (shown)$	M1 M1 (expansion of tan x, use of $\frac{1}{\sqrt{3}}$) A1

12(a)	$T = 32 - Ae^{-mt}$ $-10 = 32 - Ae^{0}$ $-10 = 32 - A$ $A = 42\ (shown)$	A1 (must see substitution)
12(b)	$T = 32 - Ae^{-mt}$ $20 = 32 - 42e^{-m(60)}$ $42e^{-60m} = 12$ $e^{-60m} = \frac{2}{7}$ $-60m = -1.25276$ $m = 0.020879$ $= 0.0209$	M1(substitution) M1(solve exponential) A1
12(c)	$22 = 32 - 42e^{-0.020879t}$ $-10 = -42e^{-0.020879t}$ $-0.020879t = \ln(\frac{10}{42})$ $t = 68.7333$ $= 69\,\text{min}\,(nearest\ \text{min})$	M1(express in ln function) A1
12(d)	$T = 32 - 42e^{-0.0209t}$ When time(t) becomes very large, $e^{-0.0209t}$ becomes very small and very close to 0. Therefore when time is large, and with the subtraction, the temperature of the chicken will close to but never reach 32°.	B1 (must explain about $e^{-0.0209t} > 0$ even when t is very large, hence temperature will always be less than 32°C)

5b


(ii) Sketch the graph of $y = c - a\cos bx$ for $0° \le x \le 360°$, indicating the turning points and end points clearly.
[3]
$y = -6\cos 2x + 3$
[1] curvature (cosine)
[1] turning points [max 9, min −3]
[1] 2 cycles.
$y = -6\cos 2x + 3$
$y = -6\cos 3x$
9
6
−3
−6
0
90°
180°
270°
360°

6(a)


11
| x | 1 | 2 | 3 | 4 | 5 |
| $\frac{y}{x}$ | 3.8 | 5.4 | 7.0 | 8.6 | 10.2 |
$y = 1.6x + 2.2$
(4, 8.6)
(1, 3.8)
$\frac{y}{x} = 6$
[1] axis/scale and table
[1] plotting
[1] st-line
2.2
2.4